

CHAPTER 10: LIGHT — MIRRORS AND LENSES

Comprehensive Class Notes, Ray Diagrams, Laws of Reflection & Optical Applications

1. NATURE OF LIGHT & SPHERICAL MIRRORS

Light Propagation: Light travels in a straight line, represented geometrically by *rays* (lines with arrows indicating direction). While plane mirrors form virtual, erect images of the same size, curved surfaces alter light paths to form enlarged, diminished, or inverted images.

What is a Spherical Mirror? A spherical mirror is a curved mirror whose reflecting surface forms part of an imaginary hollow sphere. They are created by grinding and polishing glass into a curved surface and applying a thin reflective aluminum coating.

Mirror Type	Curvature & Reflecting Surface	Image Characteristics	Practical Applications
Concave Mirror (Converging)	Curves inwards like the inner side of a spoon. Coating on outer surface.	Close object: Erect & magnified. Far object: Inverted, size varies (enlarged or diminished).	- Torch & headlight reflectors - Dentist & ENT inspection mirrors - Solar cookers (converges sunlight)
Convex Mirror (Diverging)	Curves outwards like the back side of a spoon. Coating on inner surface.	Always erect and diminished in size, regardless of object distance.	- Vehicle side-view mirrors (wide view) - Road safety mirrors at blind bends - Store surveillance mirrors

Activity 10.1 Key Insight (The Metallic Spoon Test):

- Inner Curved Surface (Concave):* Acts as a concave mirror — when held close, your face appears enlarged/erect; as moved away, the image flips **inverted**.
- Outer Bulging Surface (Convex):* Acts as a convex mirror — your face always remains **erect but smaller** in size.

2. LAWS OF REFLECTION & RAY OPTICS

When a ray of light falls upon a smooth reflecting surface, its path follows strict geometric laws valid for **all types of mirrors** (plane, concave, and convex):

- Incident Ray:** Light ray striking the mirror surface.
- Reflected Ray:** Light ray bouncing back from the mirror.
- Normal (N):** Line drawn at 90° (perpendicular) to the mirror at point of incidence (O).
- Angle of Incidence (i):** Angle between incident ray and normal.
- Angle of Reflection (r):** Angle between reflected ray and normal.

Law of Reflection	Statement & Mathematical Expression	Experimental Proof & Special Cases
First Law of Reflection	The angle of incidence is always equal to the angle of reflection: Angle (i) = Angle (r)	Measured using a comb slit, paper grid, and protractor (Activity 10.4). Special Case: If light falls along the normal (Angle i = 0°), it reflects straight back along the normal (Angle r = 0°).
Second Law of Reflection	The incident ray, the normal at the point of incidence, and the reflected ray all lie in the same plane .	Paper Bending Test (Activity 10.5): Bending the chart paper extending past the table edge breaks the plane, making the reflected ray instantly disappear.

Action of Parallel Beams on Spherical Mirrors (Activity 10.6 & 10.7):

- Concave Mirror (Converging Action):** Parallel beams fall on a concave mirror and converge at a single bright focal spot. Holding a concave mirror facing the Sun focuses solar radiation to ignite/burn paper!
- Convex Mirror (Diverging Action):** Parallel beams strike a convex mirror and spread outward in all directions.

3. LENSES — TYPES, IMAGE FORMATION & RAY BEHAVIOR

What is a Lens? A lens is a piece of transparent material (glass or clear plastic) bounded by curved surfaces. Unlike mirrors which reflect light, lenses allow light to pass through them (refraction), forming images viewed *through* the lens.

Water Drop Lens Analogy (Activity 10.8): A round drop of water placed on an oiled glass plate forms a curved transparent dome that acts as a simple convex lens, magnifying the text beneath it.

Lens Type	Physical Structure	Light Beam Behavior	Image Characteristics & Uses
Convex Lens (Converging Lens)	Thicker in the middle than at the outer edges.	Converges incoming parallel rays to a single point. Focuses sunlight to burn paper (Activity 10.11).	• Close object: Erect & enlarged (magnifying glass). • Far object: Inverted & size varies. • Uses: Spectacles, cameras, microscopes, human eye lens.
Concave Lens (Diverging Lens)	Thicker at the edges than in the middle.	Diverges (spreads outward) incoming parallel light rays.	• Always forms an erect and diminished image regardless of object distance. • Uses: Corrective eyeglasses, peepholes, optical equipment.

4. MASTER OPTICAL COMPARISON MATRIX (EXAM REVISION)

Optical Element	Light Action	Image at Small Distance	Image at Large Distance	Key Field Application
Plane Mirror	Regular Reflection	Erect, Same Size	Erect, Same Size	Dressing mirrors, periscopes
Concave Mirror	Converges Reflection	Erect, Magnified	Inverted , size varies	Torches, Dentist mirrors, Solar cookers
Convex Mirror	Diverges Reflection	Erect, Diminished	Erect, Diminished	Vehicle rear-view, Road blind spots
Convex Lens	Converges Refraction	Erect, Magnified	Inverted , size varies	Magnifying glass, Cameras, Eye lens
Concave Lens	Diverges Refraction	Erect, Diminished	Erect, Diminished	Myopia glasses, Laser optics

5. SCIENTIFIC HERITAGE, INNOVATIONS & HIGH-YIELD QUESTIONS

Scientific Heritage (Bhāskara II Era - 12th Century): Over 800 years ago, ancient Indian astronomers utilized shallow water bowls and viewing tubes aligned at precise angles to track celestial bodies. Their empirical mastery demonstrated an advanced practical understanding of the laws of reflection long before modern optical formalisms.

Modern Green Energy (Solar Cookers): In India, concave parabolic reflectors and mirror arrays are deployed in rural solar cookers to converge solar radiation onto cooking vessels, generating intense heat while conserving electricity and fossil fuels.

Question / High-Yield Problem	Detailed Answer & Scientific Explanation
Q1. Ray Angle Calculation: A light ray strikes a plane mirror such that the angle with the normal is 40°. What angle does the reflected ray make with the mirror surface?	• By the First Law of Reflection, Angle of reflection r = Angle of incidence i = 40°. • The normal makes a 90° angle with the mirror. • Therefore, the angle between the reflected ray and the mirror surface = 90° - 40° = 50°.
Q2. Normal Incidence Rule: What happens when light strikes perpendicular (normal) to a mirror?	• Angle of incidence i = 0°. • Angle of reflection r = 0°. • The light ray retraces its path straight back along the normal.
Q3. Vehicle Rear-View Assertion: Why are convex mirrors preferred as driver side-view mirrors?	• Reason: Convex mirrors curve outward, offering a dramatically wider field of view than plane mirrors and consistently forming upright (erect), diminished images of traffic behind.
Q4. Concave Mirror Museum Mystery: As a person walks towards a large concave mirror from far away, what happens to her image?	• At large distances, the image is inverted and smaller. • As she moves closer, the inverted image grows in size. • When very close, it flips to an erect and magnified image.